Leapfrogging the Melting Pot?
European Immigrants' Intergenerational Mobility Across the 20th Century

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Abstract

During the early twentieth century, industrial-era European immigrants entered the United States with lower levels of education than the U.S. average. However, empirical research has yielded unclear and inconsistent evidence about the extent and pace of their integration, leaving openings for arguments that contest the narrative that these groups experienced rapid integration and instead assert that educational deficits among lower-status groups persisted across multiple generations. Here, we advance another argument, that European immigrants may have "leapfrogged" or exceeded U.S.-born non-Hispanic white attainment by the third generation. To assess these ideas, we reconstituted three-generation families by linking individuals across the 1940 Census, years 1973, 1979, 1981-90 of the Current Population Survey, the 2000 Census, and years 2001-2017 of the American Community Survey. Results show that most European immigrant groups not only caught up with U.S.-born whites by the second generation, but surpassed them, and this advantage further increased in the third generation. This research provides a new understanding of the time to integration for 20th century European immigrant groups by showing that they integrated at a faster pace than previously thought, indicative of a process of accelerated upward mobility.

Keyword*:* intergenerational mobility, European immigrants, education, integration

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Introduction

The persistence of patterns of ethnic social distinctiveness has been a concern animating American discussions of immigration throughout the nation's history (Zolberg 2006). Such concerns played a pivotal role in shutting down the great wave of European migration that brought an estimated 15 million mostly poor, and mostly Catholic and Jewish immigrants from Southern and Eastern Europe to reside in America between about 1880 and 1920 (Benton-Cohen 2010). These anxieties seem, in retrospect, misplaced given that sociologists have generally understood rapid upward mobility coupled with widespread intermarriage to have weakened ethnic attachments and identities to the point of rendering them "symbolic" (Gans 1979) or "optional" (Waters 1990) and leading Alba to famously declare that the European ethnicities produced by the great industrial migration had reached their "twighlight" sixty years after its cessation (Alba 1981, 1985). As the narrative goes, the integrative capacities of the nation's melting pot proved overwhelming for ethnic attachments.

Yet despite the ubiquitous national influence of the melting pot story, the empirical evidence supporting this narrative remains thin, largely due to shortcomings in the nature of publicly-available census and other survey data. As we review below, while the extant results from analyses of historical census data seem to provide incontrovertible evidence of large-scale mobility between the industrial era immigrant generation and the 2nd generation consisting of their children, actual estimates of this intergenerational mobility vary considerably across studies. And perhaps more importantly, the picture of status attainment beyond the second generation remains foggy, and thus questions remain over just how long, in generational time, ethnic differences in attainment persisted. This has led to vague statements in contemporary social science theorizing that integration among the European ethnics occurred over the course of

about three or four generations, and earlier disadvantages of some European ethnic groups may even persist to the present day (Borjas 1994; Glazer and Moynihan 1970).

For example, although the narrative of the "melting pot" optimistically paints a picture of rapid upward socioeconomic mobility, at least for European origin groups, Glazer and Moynihan (1970) posited that cultural, social, and economic differences persisted among ethnic groups and integration may not have occurred until the third or fourth generation. Likewise, Borjas (1994) claimed that immigrant group disadvantages persisted into the 3rd generation and beyond. These points are important because they have been interpreted as supporting the notion that there are steep and long-lasting costs to the nation of admitting low-skilled immigrants (Borjas and Tienda 1987; Borjas 1990). Even though Borjas' findings have been called into question (Waters et al. 2007; Portes and Rumbaut 2014), his arguments continue to infuse contemporary debates about immigration policy and are used to support proposals to curtail contemporary low-skilled immigration flows (Davis and Shear 2019). If disadvantages persisted across generations for European-origin groups, wouldn't the outcome be even worse for today's immigrants, many of whom are not only low-skilled but are also likely to encounter substantial discrimination or adjustment difficulties on account of their non-European racial and cultural origins and documentation status (Portes and Zhou 1993; Waters and Jiminez 2005)?

Part of the difficulty in assessing these ideas is that data on immigrant integration are rare and incomplete. Data limitations make it especially difficult to connect immigrants' status attainments with those of their descendants three or four generations later, which makes it hard to challenge claims about the "stickiness" of European industrial-era ethnicity. Given how such claims continue to shape perspectives about contemporary immigration theory and policy (Alba,

Lutz and Vesselinov 2001), it is crucial that they be subjected to greater empirical scrutiny than they have, to date.

To that end, this paper uses newly linked U.S. Census Bureau data to provide novel empirical estimates of intergenerational progress in educational attainment among the children and grandchildren of industrial era European immigrants in the United States. We reconstitute three-generation families whose members were interviewed across multiple Census data sets spanning nearly eight decades: adults in the 2000 decennial census long form and American Community Survey (ACS) samples from 2001 to 2017, their parents in the Current Population Surveys (CPS) from the 1970s and 1980s, and their grandparents in the 1940 full-count census. These data allow us to follow immigrant families across generations and compare them with their U.S.-born contemporaries. One major benefit of our approach is that we classify groups based on the place of birth of the first generation rather than individual's ethnic identity or ancestry. This avoids potential bias related to "ethnic attrition", i.e., when members of later generations cannot be found in data due to intergenerational shifts in identity (Duncan and Trejo 2018; Alba and Islam 2009).

We focus on educational attainment as the outcome of our analyses because of its importance for socioeconomic status and position in the United States in the 20th century. Education has wide-ranging implications for an individual's socioeconomic position during his/her their adult lifetime (Haller and Portes 1973; Galobardes et al. 2006), his/her health and well-being (Kaplan, Fang, and Kirby 2017; Lynch 2003; Mirowsky and Ross 2003; Montez, Hummer, and Hayward 2012), and the life circumstances of his/her children (Dubow, Boxer, and Huesmann 2009; Erola, Jalonen, and Lehti 2016; Vollmer 2016). Industrial-era immigrants tended to arrive with lower levels of education than the U.S.-born population, so the educational

attainment of their children and grandchildren is a highly relevant indicator of structural integration into the U.S. labor force and wider society. Education provides a pathway to obtaining a job, learning English, and wage growth, and ultimately acceptance within the U.S. mainstream (Duncan & Duncan 1968; Fischer & Hout 2006). Because of the multidimensional nature of "assimilation" (see Gordon 1964) and the fact that we retain focus on one dimension of assimilation—integration into formal institutions and schools—we use the term *integration* to refer to the process whereby immigrants and their descendants experience equal opportunity within U.S. institutions and, eventually, attain similar outcomes as the U.S. mainstream, often proxied by U.S.-born whites (Alba 2003). We also utilize the term *mobility* to denote change in educational attainment across generations.

We find that many European immigrants had lower levels of educational attainment than their U.S.-born white counterparts. However, these deficits were minimized or eliminated by the second generation descendents of immigrants from most European nations. Moreover, by the third generation, the grandchildren of immigrants had attained even higher levels of education than the descendants of U.S.-born whites. As we discuss, the results help put to rest controversies about the persistence of disadvantage among European-origin groups and speak to the degree and speed to which immigrants were able to integrate. However, the results also leave open questions for further research about whether the experiences of European-origin immigrants can be applied to non-European groups, a point to which we return in the article's conclusion.

Theoretical Expectations

Upon arrival to the U.S. in the early 20th century, European groups differed dramatically from one another and from the U.S.-born with regard to the amount of social and economic

capital they possessed. Alba (1990) characterizes European immigrant groups as fitting within a status hierarchy whereby Protestant groups from the British Isles occupied the top (i.e., the highest attaining status group), other groups from Northern and Western Europe occupied the middle, and at the bottom were the Southern and Eastern Europeans. Northern and Western Europeans (NWE) such as the German and English, were mainly literate, skilled, and quick to assimilate. Southern and Central European (SCE) migrants such as Italians, Jews, Poles, and others from Balkan regions were predominately labor seeking, occupying industrial jobs in manufacturing, construction, and steel industries (Hutchinson 1965). These status attainments and positions within the hierarchy differentiated each of these ethnic groups by place of origin. As historical accounts of discrimination show, Irish, Jews, and Italians alike were targets for nativist hostility (Alba and Nee 2003; Foner 2000; Higham 1955). This discrimination and separation of classes meant that the opportunity structures among immigrants were also different. Immigrants often lived and worked in separate spheres which may have limited the educational opportunities for certain groups, at least prior to educational expansion in the 1950s and before major changes in compulsory schooling (Sassler 2006; Alba 1985).

It remains unclear whether and how quickly these initial disparities among European ethnic groups disappeared. We anticipate three possibilities. The first predicts *persistent ethnic disadvantage*. According to the status attainment model, parental status (measured here as educational attainment) is the strongest predictor of a person's attainment (i.e., education) in adulthood, leading to the reproduction of a family's class position across generations (Blau and Duncan 1967; Haller and Portes 1973). Immigrant starting points are also thought of as "class origins" (or class positions) and are representive of initial status attainments of the first immigrant generation, which are measured here as educational attainment. Indeed, many studies

in the United States show that even as absolute levels of education have increased, there has been little change in relative class position (i.e., educational attainment given European national origin) for families over time (Bowles, Gintis, and Osborne Groves 2005). Moreover, some earlier European immigrant groups may have faced discrimination and other barriers to upward mobility which in turn could lead to some groups rejecting schooling and legitimate pathways to success (Alba and Nee 2003; Feliciano 2006; Portes and Zhou 1993). Indeed, not all immigrant groups were supportive of mandatory schooling and preferred that their children worked or helped with other family matters at home, and some even turned to criminal activities and organizations (e.g., Italians; see Alba 1985; Pearlmann 2005). This line of argumentation, which we label as the *persistent disadvantage model*, is similar to that predicted by segmented assimilation theory, whereby groups arriving with few resources and encountering unsupportive contexts of receptions are predicted to experience stagnant or slow assimilation patterns (Portes and Zhou 1993). Consistent with these ideas, Borjas (1994) argued that human-capital differences among immigrant groups at the beginning of the 20th century were still evident among third-generation descendants three-quarters of a century later, though in muted form. His study found that the relative ordering of national origin and ancestry groups among individuals was similar in the 1910, 1940, and 1980 censuses, and concluded that it may take a century or more until differences among immigrant groups disappear. However, as we explain in the next section, the empirical evidence for these ideas is inconclusive due to data limitations that make it difficult to link immigrants' origins with outcomes of their descendants.

This leads us to a second possibility that predicts *integration* whereby initial ethnic differences quickly attenuate across generations. There are several reasons to expect European immigrants to have rapidly closed the gaps in education attainments relative to U.S.-born whites.

First, European immigrants were well-positioned due to their geographic concentration in the northeast region of the United States and their place in history to take advantage of the educational reforms and expansions occurring in the United States in the first half of the 20th century. In the early 20th century, educational reforms had introduced compulsory attendance laws that increased the access and opportunity for individuals to obtain an education (Katz 1976; Lleras-Muney and Shertzer 2015; Richardson 1980). These improvements were especially pronounced in the northeast—the geographic locations often inhabited by industrial-era European immigrants.

Additionally, the economic growth and expansion in higher education that occured during this time period, coupled with large federal investments in the opportunity structure, permitted European immigrants and their children to achieve upward mobility (Alba 2020; Lieberson 1980). As a part of the New and Fair Deals enacted as an attempt to improve economic conditions that resulted from the Great Depression and World War II, the Servicemen's Readjustment Act of 1944 (i.e., the GI bill) was introduced to create enhanced opportunity for college attendance, after which a college degree became more commonplace (Kim and Rury 2007). As argued by Alba (2020), the rapid expansion in schooling and the labor market provided the structural changes that enabled a non-zero-sum pattern of assimilation, meaning that European-origin immigrant groups could advance without having to out-compete or displace others located above them in the status hierarchy. In other words, the expansion in educational opportunities and compulsory school attendance laws reduced inequalities in educational attainment (Alba 2020). These ideas lead us to hypothesize that differences in educational attainment between the descendants of U.S.-born whites and the descendants of immigrants will be reduced to nonsignificance across generations, hereby discussed as the *rapid integration*

model. Indeed, historical studies have found that immigrant groups improved their educational status over time and between the first and second generations (see White and Mullen, 2016; Sassler, 2006).

A third possibility predicts *accelerated upward mobility,* whereby the descendants of European origin immigrants not only reached equal attainment levels as the U.S. mainstream, but surpassed them, thus exceeding the expectations of the status attainment model. In other words, we hypothesize that the descendants of immigrants "leapfrog" the descendents of U.S.-born whites across generations. One reason to anticipate this *accelerated upward mobility model* among immigrant families is that immigrants were able to draw upon co-ethnic and familial experience and resources to assist in their settlement and integration. The immigrant communities that European groups were settling in provided enclaves where immigrants could live, work, and seek support from earlier-arriving co-ethnics (Alba and Nee 2003; Borjas 2000). White and Mullen (2016) found that more established and early-arriving immigrant groups (e.g. Northern and Western Europeans) improved their socioeconomic status to a lesser extent than did Italian and Polish groups, suggesting that later-arriving groups were given a hand-up by earlier-arriving co-ethnics, enabling them to "catch up" more quickly than similarly low-educated U.S.-born counterparts. One way this may have occurred is through the rise in Catholic educational enrollment (Baker 1999). To improve their education, the Catholic working class formed their own schools, which paralleled the rigor of the public schools' nonreligious curriculum (Baker 1992), yet allowed them to be culturally accommodating (Gleason 1970; Lazerson 1977; Moore 1986). By adapting a model of mass schooling from the public sphere, Catholic schools were able to educate some of the least-educated populations in U.S. society (Baker 1992).

Another reason to expect accelerated mobility among immigrant families relates to the fact that some immigrant groups lacked occupational mobility ladders and faced barriers within the mainstream business world (Catron 2016). Furthermore, occupational structures changed dramatically between 1930 and 1970 (Alba 2020). Even if they wanted to, people could not have followed their immigrant parents or grandparents into similar occupations as these jobs grew scarce; education became required to obtain the white-collar jobs that were proliferating. As these shifts in occupational structure occurred, pathways to improving a family's social position were rerouted through education. The education route to upward mobility may have been particularly important for European ethnics who were not yet fully accepted, such as Italian Americans (Alba 1985).

A third reason to anticipate a pattern of accelerated upward mobility is that children of immigrants often feel obligated to fulfill their parents' dreams for upward mobility in return for the sacrifices they made to move to the United States. Fulfilling such obligations could mean working long hours to contribute to household income, but it could also mean pursuing secondary and postsecondary educational credentials, particularly in circumstances in which higher education is available, affordable, and is recognized as having clear financial benefits. As noted above, such circumstances started to emerge especially for men during economic and educational expansion in the decades following World War II. Children of immigrants may have felt more pressure than children of U.S.-born parents to take advantage of these expanding educational opportunities.

Indeed, pressure to succeed has been observed among present-day immigrant families. For example, Kao and Tienda (1995) suggest that some of the characteristics often found in immigrants, such as their optimism to search for new opportunity via migration, are responsible

for their increased success. Lee and Zhou (2015) found that Asian immigrant families promote educational success through family values that place high emphasis on educational performance. Another study found that, due in large part to positive selection processes, immigrant families tend to have high expectations for their children's education, and implement strategies to promote their children's educational achievement (Feliciano and Lanuza 2017). Although these ideas have not been applied to industrial-era European immigrants, we know, for example, that Jewish immigrants placed great importance on education in efforts to get ahead (Gibson and Ogbu 1991). Although not all European ethnic groups were as focused on eduation as were Jewish immigrants, it is possible that the positive selection processes and the strategies observed among today's immigrants were also evident among early 20th century European immigrants, particularly during the post-WWII period when educational opportunities expanded rapidly and routes to the middle class became increasingly tied to educational attainment.

Overall, due in large part to severe data limitations, there is no clear consensus about the degree to which initial differences in immigrant groups persist over generations. Some have advanced the *persistent disadvantage* model, such as Borjas (1994), who argued that ethnicity is "sticky", and it takes several generations to reach parity with U.S.-born whites. Others have advanced a *rapid integration* model, such as Alba and Nee (1997, 2003), who argue that the rapid expansion of opportunities in education made early 20th century European groups' rapid integration into the U.S. mainstream possible. Still others have advanced the *accelerated upward mobility* model (e.g., Feliciano and Lanuza 2017), but to our knowledge, few have applied this idea to early 20th century European immigrant groups.

Empirical Evidence of European Ethnic Change across Generations

Until recently, no large-scale data source existed to observe inter-generational progress across immigrant generations *within* European immigrant ethnic families (from first generation grandparents to second-generation children, to third-generation grandchildren, and so on). Thus, the production of empirical evidence germane to questions of the persistence (or absence) of ethnic disadvantage among the offspring of European immigrants has relied on less-direct observational methods–typically either synthetic cohort analyses of the various immigrant generations as they appear in multiple survey cross-sections over time (e.g., Borjas 1994; Alba, Lutz and Vessilov 2001), or comparisons of integration outcomes (such as educational attainment) across immigrant generations or birth cohorts observed within a single survey cross-section (e.g., Alba 1981, 1985; Hirschman and Kraly 1990; Sassler 2006; Neidert and Farley 1985). The result is an empirical literature that presents a rather overwhelming, if blurry and incomplete, picture of ethnic convergence to levels of outcomes exhibited by whites of "native stock", defined as white children with US-born grandparents (the "4th-plus-generation") (Sassler 2006). *En masse,* the children and grandchildren of European immigrants of the industrial era seem clearly to have converged to average attainment levels of the native-stock population. However, the extent to which the offspring of European immigrants may have exceeded native attainment levels, the number of immigrant generations required to realize convergence, and whether the integration trajectories and intergenerational mobility patterns vary across ethnic groups, are questions without clear answers owing primarily to data and methodological limitations inherent in the established body of empirical research, which we review here.

Analyses of data collected during or in the decades following industrial-era migration demonstrate that the process of intergenerational mobility toward attainment-convergence with U.S.-born whites was already well underway. Sassler's (2006) analysis of 1920 census data revealed that, even before immigration to America from around the world was largely curtailed by restrictive entry laws enacted in the 1920s, the U.S.-born (second-generation) children of nearly all the major industrial-era European ethnic groups made clear gains in their rate of school enrollment, over that of their European-born (first generation) peers, even if their rates failed to converge to U.S.-born white levels. For example, among the native stock, 53 percent of adolescent boys and 62 percent of adolescent girls (aged 15-18) were enrolled in school at the start of the 1920s. Among Italian- and Polish-origin girls, there was no improvement in the enrollment rate from the first to the second generation with rates between 20 and 30 percent, but their co-ethnic male peers made modest to substantial gains between the first and second generations. The increase was most notable for Italian boys among whom the enrolled share climbed from 28 percent to 37 percent from the first to the second generation. In a theme echoed throughout the empirical literature, Jews stand apart from other industrial-era ethnic groups for their relatively high levels of attainment; 42 percent of first-generation Jewish boys and girls were enrolled in school in 1920, and made gains toward parity with U.S.-born whites as enrollment rates increased to 48 and 46 percent among second-generation boys and girls, respectively (Sassler 2006, p.10, Table 2).

The patterns of ethnic and generational variation in adolescent school enrollment observed in Sassler's (2006) analysis of 1920 Census data are echoed in the educational attainments observed for these adolescents when they were adults in the middle of the century. Hirschman and Kraly (1990) present statistics from 1940 and 1950 Census data showing large

deficits in average years of schooling completed among working aged immigrant men from major source countries of industrial-era migration relative to the U.S. born. Unlike in the case of second-generation school enrollment among adolescents in 1920, however, U.S.-born men of industrial-era ethnic groups (who in 1950 would have been overwhelmingly of the second generation) reported average educational attainments that were equal to those of the U.S.-born majority; men born in America to Russian or Ukrainian parents (a group that was predominantly Jewish) again proved exceptional, opening up an average schooling advantage of two years over U.S.-born whites (Hirschman and Kraly 1990, Table 3). Thus, it appears that by the middle of the twentieth century, the second-generation children of the industrial-era immigrants had either matched or exceded the attainments observed for U.S.-born whites.

Evidence about intergenerational mobility, which requires information about the attainments of both children and their parents (and grandparents), however, has remained murky. Samples collected in the ensuing decades and analyzed by sociologists, including a 1962 Current Population Survey (CPS) module focused on inter-generational mobility (Duncan and Duncan 1968), analyses of the 1979 CPS, which first introduced a question about persons' ancestry (Alba 1985; Neidert and Farley 1985), and samples from the 1970s General Social Surveys (GSS) (Alba 1981), all show general patterns consistent with dramatic reductions in the attainment gap between U.S.-born whites and members of industrial-era ethnic groups across immigrant generations. Nevertheless, the evidence accumulated in the decades following the industrial-era migration, while suggestive of substantial group mobility toward mainstream convergence, lacks the precision necessary to confidently make claims about the pace at which convergence occurred (assuming it did) and the degree to which it varied across the major ethnic groups migrating to America during the industrial era. These data limitations, which relate to the

inability to confidently define an individual's generational status and national origins, are exemplified in a long-standing debate between George Borjas and Richard Alba.

In an influential paper, noted above, Borjas (1994) casts the issue of the persistence of industrial-era ethnic skill differentials as an open empirical question. Curiously, despite drawing considerably from the debates in sociology in the latter decades of the 20th century over the theory of the melting pot, Borjas cites none of the empirical evidence (summarized above) produced by sociologists over the same time period. In order to estimate the generational length over which ethnic skill differentials might persist, Borjas calculates ethnic group-level estimates of human capital upon arrival in the United States–-–-approximated by literacy rates–-–-using data from the 1910 Census. He then regresses the educational and occupational attainment of second-generation cohorts observed in the 1940 census, and grouped into immigrant ethnicities based on reported parental place of birth. Similarly, Borjas identifies third-generation European ethnics in the 1980 census, using a newly introduced ancestry question to approximate the original 1910 immigrant ethnic groups and then regresses third-generation attainment on the group-level literacy rates in 1910. Most pertinent to the discussion here, Borjas finds that the group-level literacy rates in 1910 are significant predictors of educational attainment even into the third generation. On the basis of these results, he concludes that ethnic differences in human capital observed among immigrants are still observable in U.S.-born generations nearly a century later.

The robustness of Borjas's (1994) conclusions were called into serious question in a replication study performed by Alba et al. (2001). Using GSS data in combination with Census data, they demonstrate that the effects of group-level human capital in the immigrant generation on educational attainment in the third generation are dependent largely on Borjas's seemingly

questionable decisions related to the measurement of both ethnicity and immigrant generation. Alba and his colleagues (2001) present evidence suggesting massive inconsistencies in how ethnic groups are measured over time. For example, Borjas (1994) identifies the first generation based on adults' reported country of birth in 1910, the second generation based on adults' reported parents' country (or countries) of birth, but the third and later generations are defined based on reports of persons' *ancestry* using a Census question introduced in 1980. The problem with this approach can be seen in the example of a hypothetical third-generation individual whose grandparents were Jewish immigrants from Austria (thus identified as "Austrian" in Borjas's design), but who identifies their (third generation) ancestry as "Jewish" or "Yiddish". Similarly, the descendants of German-speaking immigrants from Russia may well identify ancestrally with their grandparents' mother tongue (German) rather than as Russian immigrants of the industrial era. Such "slippage" in ethnic measurement over the immigrant generations makes research designs like the one employed by Borjas susceptible to large potential biases, yet it is also impossible to know the extent and direction of such bias in the absence of better data.

Alba et al. (2001) highlight similar inconsistencies attending Borjas's measurement of immigrant generations, specifically the third (grandchild) generation. In a cross-sectional sample, decennial census and survey measures of immigrant generations become imprecise after the child (second) generation because surveys do not probe for immigrant origins extending into the past beyond the parent generation. This creates an amorphous "third+-generation" (Bean et al. 2015) that lumps the grandchildren, great-grandchildren, and so on, into a residual generation with unobservable heterogeneity. In the case of the Borjas study, Alba and his colleagues (2001) point out that the "3rd generation" consists of any U.S.-born adult in the 1980 decennial census reporting a given ethnic ancestry. The fundamental assumption in this approach is that U.S.-born

adults expressing a given ethnic ancestry in 1980 are overwhelmingly the grandchildren of industrial-era European immigrants, which Alba et al. (2001) show to be "demonstrably false" in the cases of some ethnic groups, most notably for Italian Americans.

If given equal weight with similarly-designed studies of ethnic persistence, the conclusion drawn by Borjas--that ethnic disadvantages originating with low-skilled European immigration have persisted for a century--stands apart as a minority interpretation of extant census data. Nevertheless, this and similar work by Borjas, has stoked long-standing concerns about the societal and economic impacts of immigration, including the undermining of collective national identity and social cohesion (e.g., Huntington 2004; Putnam 2007). Its role in shaping conservative approaches to managing immigration seems to have far outstripped its influence in social science research on immigration in the academy (see, e.g., Douthat [2013]). The existence of blind-spot producing methodological limitations lend to the overall perception of a long-standing unresolved debate about patterns of integration and ethnic persistence among the offspring of industrial era European immigrants, and provide an opening for alternative interpretations that paint the U.S. immigration experiment as imposing large net costs to society. As we explain next, we leveraged newly-available historical data linkages to help bring clarity to these debates.

Data and Methods

Sample

The data for this study are drawn from the full-count 1940 census, years 1973, 1979 and 1981-1990 of the Current Population Survey (CPS), the 2000 census long form, and years 2001-

2017 of the American Community Survey (ACS)[1]. The 1940 Census was the first U.S. census to collect data on educational attainment[2].

The Census Bureau attached unique personal identification keys (PIKs) to individuals in the 1940 census and CPS data files as part of its Census Longitudinal Infrastructure Project (CLIP), which enabled us to link individuals and families across data sources (Massey et al. 2018). Contemporary census data and surveys include sufficient personally identifiable information (PII) such as name, exact date of birth and address, that when combined with administrative records from the Social Security Administration, Internal Revenue Service, and other agencies, enables the Census Bureau to accurately assign PIKs to the vast majority (90% or more) of respondents[3]. Until recently, limited PII and administrative data has restricted the share of individuals in historical census data that can be assigned a PIK. For example, many historical studies that used linked census data were restricted to individuals whose names do not change over time, and who have a unique name and year of birth combination. As a result, historical studies have tended to have low linkage rates (typically around 20%), and they tended to exclude women, whose names often changed with marriage. However, the CLIP modified the linkage methodology to substantially increase the share of 1940 Census responsdents that could be assigned a PIK based on name, age, state or country of birth, state of residence in 1940, state of residence in 1935, and parents' names, and this information could in turn be linked to individual's Social Security Administration record even if they did not enroll in the Social Security System until years later (Massey et al. 2018). The additional matching variables enabled

[1] A gap in data years is the result of unavailable data (i.e., 1950, 1960 or 1970 census long-form), where PIKs have not yet been assigned.

[2] We analyzed these restricted data in the Penn State Federal Statistical Research Data Center, and all results presented here were disclosed for review by the Census Bureau Disclosure Review Board. CBDRB-FY21-039 Approved on 11/2/20 and CBDRB-FY2021-CES005-019 approved on 6/11/21.

[3] For more information on the linking process, see Wagner and Layne (2014).

the Census Bureau to assign PIKs to 46% of individuals in the 1940 Census, and to 73% of those aged 0-19—the group primarily included in our analysis.

We used the CLIP data to connect members of families who were linked in data collected during three time periods, resulting in a three-generation family file. Parents with co-residential children were identified in the 1940 census (the "first family generation"). Then, the co-residential children of those parents were identified and linked to their later adult records in the CPS (the "second family generation"). The co-residential children of these individuals were located and linked to their adult records in the 2000 census and/or the 2001-2017 ACS (the "third family generation"). We limited the sample to two groups: (1) families in which at least one of the grandparents in the first family generation was born in Europe and none were immigrants from other world regions, and (2) families in which all of the grandparents in the first family generation were born in the United States and identified as non-Hispanic white. Thus, our study focuses on comparisons of European immigrant families with each other and with a contemporaneous sample of non-Hispanic U.S.-born whites, a group we interpret as representing the U.S. mainstream in 1940 and the post-WWII period. Such comparisons help us gauge the relative progress observed for European immigrant families within the context of the rapid rise in educational expectations and opportunities that occurred for non-Hispanic whites throughout the twentieth century. The considerable differences in civil rights and educational opportunities between European immigrant and non-white racial-ethnic and national origin groups (e.g., African-Americans and Mexicans) require more theoretical development than we have space for here. We thus limit our comparison group to non-Hispanic U.S.-born whites.

We organize the data from the perspective of adults aged 25 or older in the third family generation, as shown in Figure 1. About 11,000 third family generation adults (63%) are linked

to their mothers and at least one maternal grandparent, with 93% of this group linked to both maternal grandparents. Additionally, about 12,500 third family generation adults (71%) are linked to their fathers and at least one paternal grandparent, with 93% of this group linked to both paternal grandparents. These two groups overlap; about one-third of third-generation adults are linked to both their mother's and father's sides of the family. Altogether, the sample includes approximately 17,500 third family generation adults age 25 or older who are linked to at least one parent and at least one grandparent (8,500 women and 9,000 men age 25+).

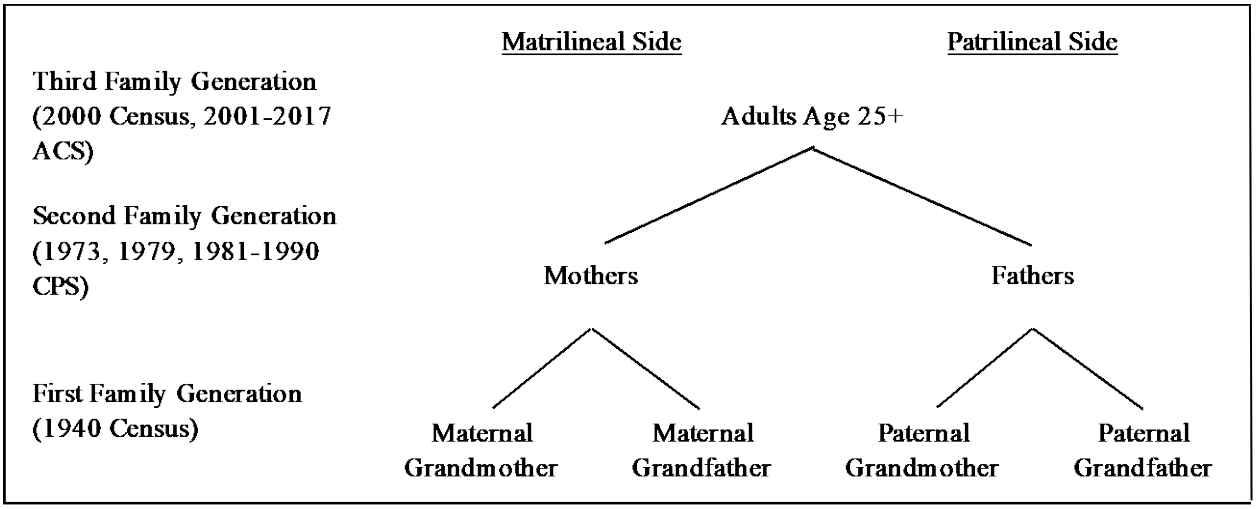


Figure 1. Family Members in Sample (from perspective of contemporary adults)

Linkage Bias

Although a large number of individuals are assigned a PIK, not everyone is. The rate at which individuals are linked largely depends on the quality of information available to identify them, which varies by data source (Leach, Van Hook, and Bachmeier 2018). The rates of PIK assignment are 46 percent in the full 1940 census but 73 percent for those aged 0-19, 71 percent in the CPS, and over 90 percent in the Census 2000 and ACS files. Moreover, some groups have higher PIK assignment rates than others (Bond, Brown, Luque, and O'Hara 2014; Wagner and Layne 2014), and immigrants have been shown to have lower linkage rates than the U.S.-born,

presumably because of their lower levels of representation in administrative records (Bond, Brown, Luque, and O'Hara 2014). Luckily, the construction of our sample does not rely on the linkages of foreign-born individuals. Instead, we link the U.S.-born children of immigrants in the 1940 Census to their adult records in the CPS, and then we link their children (the grandchildren of immigrants) in the CPS to their adult records in the 2000 Census and 2001-2017 ACS. Nevertheless, there may still remain other biases due to differential rates of being assigned a PIK. Additionally, our sample captures only families with children at given points in time (i.e., parents with children present in the household in 1940 Census and grandchildren present in the household of their adult children between 1972 and 1990).

To reduce the impact of these potential sources of bias, we developed and use inverse probability treatment (IPT) weights, following the methods of Hong and Raudenbush (2008) and Sampson, Sharkey, and Rudenbush (2008). To calculate the IPT weights, we first model the probability that second family generation children (observed in the 1940 Census) are followed up as adults and have co-residential children in the CPS (the third family generation child). We then model the probability that the third family generation children are linked as adults in the 2000 Census or the ACS. Predictors include age, race, national origin, and years of parental education. For the final IPT weight, we multiply the predicted probability of selection into the second and third generation samples together and take the reciprocal. The IPT weights are then normalized so that the final weights sum to the number of cases in the analytic sample. The adjusted weights give families who are least likely to be included in the sample (given observed characteristics) greater influence in our analysis. In practice, the application of the weights shifts the distributions of age and education of the grandparents slightly downward compared with the unweighted sample, but results regarding educational mobility are nearly identical regardless of

whether or not weights are applied.

Measures

A key organizing variable in the analysis is family generation. As noted earlier, the parents found in the 1940 Census compose the first family generation, their children compose the second family generation, and their grandchildren make up the third family generation. To provide some historical context, the average year of birth for the first family generation was 1902 (SD = 8.8); they attended school primarily during the first three decades of the twentieth century. The second family generation was born on average in 1932 (SD = 6.1), and they attended school primarily during the 1930s, 1940s, and 1950s. Finally, the third family generation was born on average in 1963 (SD = 6.4); they attended school primarily during the 1960s, 1970s and 1980s.

The educational attainment of various family members serve as both outcomes and predictors in our analyses. We measured educational attainment as the highest grade of education attained for adults age 25 or older in the third family generation, and their mothers, fathers, maternal grandmothers, maternal grandfathers, paternal grandmothers, and paternal grandfathers. In analyses of educational mobility between the first and second family generations, we treated educational attainment of the adults in the second family generation as the outcomes, and the average educational attainment of their parents (i.e., the maternal grandparents and paternal grandparents in the first family generation) as predictors. In analyses of educational mobility between the second and third family generations, we treated the educational attainment of the adults in the third family generation as the outcome, and the average attainments of grandparents and parents in the first and second family generations, respectively, as predictors. When employed as independent variables, we average the educational attainments of the grandparents

and parents in the first and second family generations to reduce collinearity. In supplemental analyses, we tested alternative ways of summarizing educational attainment (e.g., highest, father's/grandfather's, mother's/grandmother's education), and the results were very similar to those presented here. For the second generation, in models that separate results by sex (e.g., Tables 2 and 3), we follow the matrilineal (all women) or the patrilineal (all men) sides of the family. When separating results by sex for the third generation, we include both maternal and paternal sides of the family (i.e., "combined" models), which we explain further below.

Ethnic origin is a family-level measure based on the place of birth and racial-ethnic identification of the grandparents in each family. Ethnic origin is treated as fixed across generations even if their children or grandchildren no longer self-identify with the ethnic group. This is an important strength of our study because it makes it possible to assess the intergenerational mobility of immigrant groups regardless of the level of inter-ethnic marriage or changes in identification (Alba 2020; Duncan and Trejo 2018), factors that are likely to confound the relationship between generational status and educational attainment.

We created separate ethnic origin measures for the matrilineal and patrilineal sides of the family, with the former based on the place of birth and racial-ethnic identity of the maternal grandparents in the first family generation and the latter on the paternal grandparents in the first family generation. We also created a combined measure of ethnic origin for each family based on the origins of both sides of the family. For the matrilineal, patrilineal, and combined measures of ethnic origin, we distinguished among families in which the grandparents in the first family generation were born in Ireland, other parts of Northwest Europe, Central Europe, Eastern Europe, and Southern Europe.[4] We separated Ireland from the rest of Northwest Europe because

[4] We employ the country region classifications defined by IPUMS for 1940 (https://usa.ipums.org/usa-action/variables/BPL#codes_section).

of its large group size and unique migration history. The reference group includes families in which both grandparents in the first family generation were non-Hispanic white and born in the United States. For convenience, we refer to the first five groups as "European immigrant families" and the last group as "U.S.-born white families". Roughly 80% of the families across the linked datasets were U.S.-born white families, while the remaining were distributed across the five European origin groups (see Table 1).

The ethnic origin of grandparents in the first family generation is not uniform within families. In about 25% of European immigrant families, one of the grandparents was born in the United States while the other was born in Europe. In such cases, we assigned the ethnic origin of the immigrant grandparent. For 9 percent of the families in which both grandparents were European immigrants, the grandmother and grandfather were born in different regions of Europe. In these cases, we assigned the ethnic origin in the following order of priority: Ireland, Northwest Europe, Southern Europe, Central Europe, and Eastern Europe. We followed the same decision rules to create the matrilineal, patrilineal, and combined measures of ethnic origin; for the combined measure, we took into consideration the place of birth and racial-ethnic identity of all four grandparents rather than just the two maternal grandparents or the two paternal grandparents.

Finally, in regression analyses we accounted for a number of demographic factors: year of birth (continuous), region of residence in childhood (Northeast, South, Midwest, and West), household size in childhood (continuous), whether the person lived with two parents in childhood, and year of the CPS survey. For the analyses of second-generation attainments (based

Table 1: Sample Descriptives (mean or proportion; SD shown for continuous variables only)

	Matrilineal Side		Patrilineal Side		Combined (Both Sides)	
	Mean	SD	Mean	SD	Mean	SD
Third Family Generation (2000 Census & 2001-2017 ACS)						
Years of Education	14.2	2.5	14.2	2.4	14.1	2.5
Year of Birth	1962.0	6.2	1963.0	6.2	1963.0	6.4
Age	45.0	8.8	43.7	8.7	44.1	8.8
Male (proportion)	0.51		0.51		0.51	
Household Size in Childhood	4.70	1.73	4.85	1.62	4.73	1.66
Region in Childhood (proportion)						
Northeast	0.30		0.29		0.29	
Midwest	0.39		0.39		0.38	
South	0.20		0.21		0.21	
West	0.11		0.11		0.11	
Two parents in Childhood	0.58	0.49	0.50	0.50	0.37	0.48
Ethnic Origin (proportion)						
US-born NH-White	0.80		0.80		0.76	
Northwest Europe	0.06		0.07		0.08	
Ireland	0.02		0.01		0.02	
Eastern Europe	0.02		0.02		0.03	
Central Europe	0.05		0.05		0.06	
Southern Europe	0.05		0.05		0.06	
Second Family Generation (CPS)						
Years of Education	12.4	2.3	12.8	3.1	12.6	2.7
Year of Birth	1933.0	5.8	1932.0	6.4	1932.0	6.1
Age	47.0	7.3	48.4	7.6	47.7	7.5
Household Size in Childhood	5.67	2.15	5.70	2.24	5.68	2.01

Continued on next page.

Table 1 (Continued): Sample Descriptives (mean or proportion; SD shown for continuous variables only)

	Matrilineal Side		**Patrilineal Side**		**Combined (Both Sides)**	
	Mean	SD	Mean	SD	Mean	SD
Region in Childhood (proportion)						
Northeast	0.31		0.30		0.30	
Midwest	0.40		0.41		0.39	
South	0.20		0.20		0.21	
West	0.09		0.09		0.09	
Two parents	0.9	0.3	0.9	0.3	1.0	0.2
CPS year	1980.0	5.5	1980.0	5.5	1980.0	5.5
First Family Generation (1940 Census)						
Years of Education	8.6	3.1	8.4	3.1	8.5	2.9
Year of Birth	1903.0	9.0	1902.0	9.4	1902.0	8.8
Age	37.4	9.0	38.4	9.4	37.8	8.8
N	11,000		12,500		17,500	

Notes: IGENS-20 Sample, Vintage 1 (3-generation families; 1940 Census; 1973, 1979, 1981-1990 CPS; Census 2000; 2001-2017 ACS). Approved for disclosure on 11/2/20; review number: CBDRB-FY21-039. Because of rounding rules set forth by the Census Bureau Disclosure Review Board, not all column proportions may add up to 1.

on the CPS samples), we include sample-year indicators to account for any secular increases in educational attainment that may have occurred during the time period covered in the CPS samples, 1973-1990. Means for all analytic variables are presented in Table 1.

Analysis

Because women's opportunities in schooling and the labor force were more constrained than men's, especially before the 1970s (Fischer and Hout 2006), we conducted all analyses separately for women and men. Ethnicity is defined using the matrilineal measure for women and the patrilineal measure for men.

We first compared mean education levels across generations by ethnic origin to assess changes in ethnic group differences in educational attainment (Figures 2 and 3). We followed this descriptive analysis with regression models that assess whether the descriptive findings hold after adjusting for demographic factors (Table 2). Disadvantages for European immigrant families relative to U.S.-born white families, especially if observed as late as the third family generation, would provide evidence of persistent disadvantage. Small and nonsignificant differences for the second and, especially, the third family generation would support the integration model. Finally, advantages of second- and third-generation European immigrant families relative to U.S.-born white families would provide evidence of accelerated upward mobility.

Second, we assessed the extent to which the second- and third-generation disadvantages or advantages are explained by differences in intergenerational mobility as opposed to family class origins or "starting points". To do this, we controlled for parents' educational attainment. In models predicting the third generation's attainment, we controlled for both parents' and grandparents' educational attainment (Table 3). If the ethnic origin coefficients attenuate toward

zero, this would suggest that starting points, as opposed to inequalities in the mobility process, explain group differences. However, if ethnic differences persist or even increase, this would reflect inequality in opportunities to advance beyond the educational attainments of one's parents and grandparents.

Third, we investigated heterogeneity in the mobility process. We were particularly interested in whether European immigrants' disadvantages compounded or were compensated for in subsequent generations, so we tested interactions between parent education and ethnic origin. To interpret the results, we generated and graphed predicated values for significant interaction effects, while holding all controls at their mean levels (Figures 4 and 5).

Finally, we recognize that the assignment of ethnic origin was somewhat arbitrary, particularly for the third generation. There was little interethnic (and virtually no interracial) marriage among maternal and paternal grandparents in the first family generation, meaning that ethnic origin was relatively unambiguous for them and their second generation children. However, their second generation children were much more likely to marry outside their ethnic group, so many of the grandchildren of the original immigrants have equal claim to two different ethnic origin groups, one on their mother's side of the family and another on their father's side. For this reason, we run the analyses of third generation educational attainment using all three versions of ethnic origin: the matrilineal, patrilineal, and combined measures (Appendix Table 3).

Results

Educational Attainment across Generations

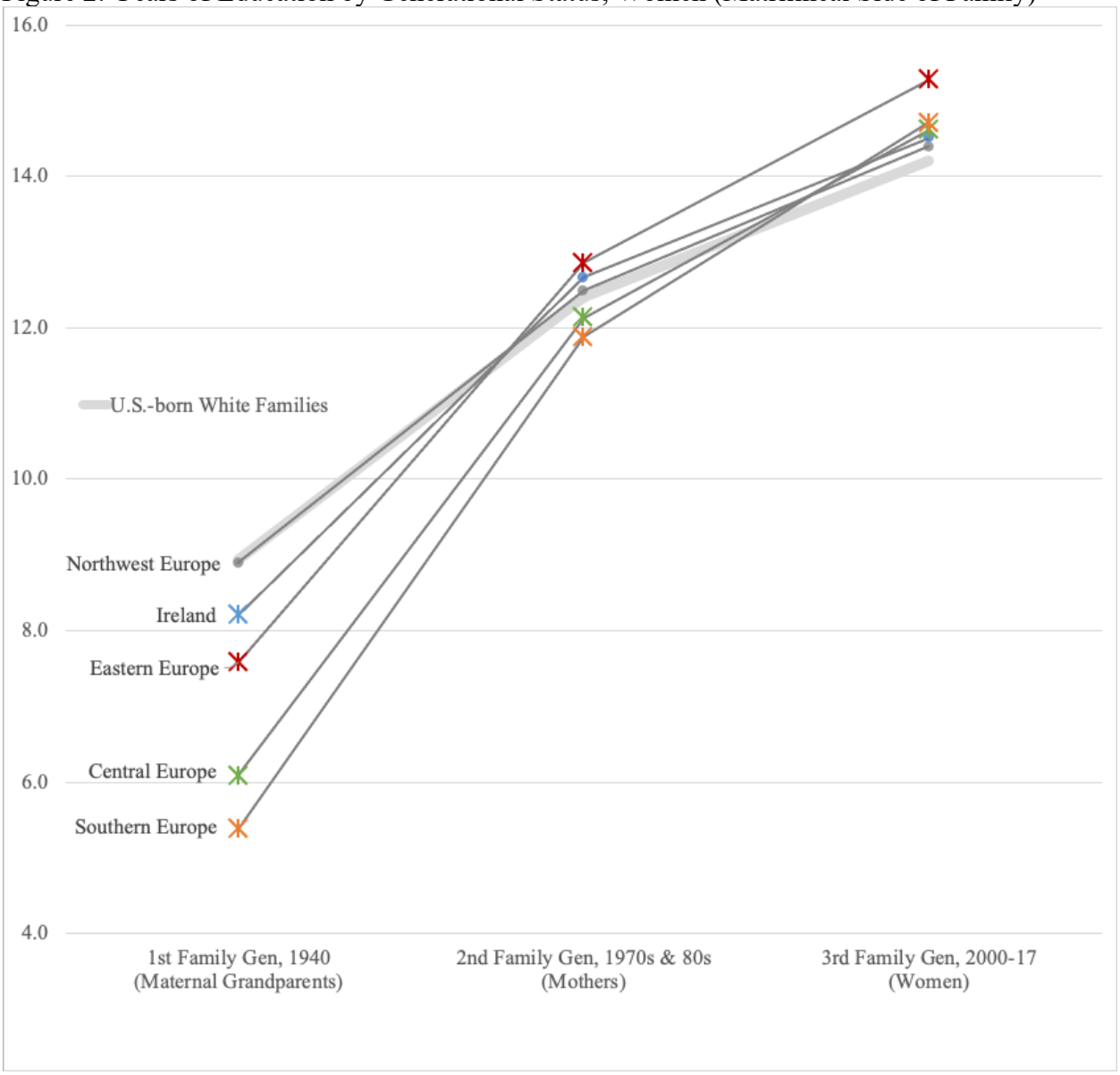
To what extent did differences in educational attainment between immigrant groups and

U.S.-born whites change across generations? To answer this question, we display the average years of education for women by generation (first-generation maternal grandparents, second-generation mothers, and third-generation women) in Figure 2, and the same information for men (first-generation paternal grandparents, second-generation fathers, and third-generation men) in Figure 3. Ethnic origin is based on the origins of maternal grandparents for women and paternal grandparents for men. Significant differences between the European immigrant origin and U.S.-born white families are indicated by the "X"-shaped markers in the graphs (with small circles indicating nonsignificance).

In 1940, first-generation European immigrants, with only a few exceptions, had significantly lower levels of educational attainment than their contemporary U.S.-born white counterparts. Additionally, there was large variation among the different European immigrant groups. Northwestern European and Irish immigrants tended to have the highest levels of education, on par with U.S.-born whites; in fact, there were no significant differences from U.S.-born whites among Northwestern European women and Irish men. Eastern European immigrants fell in the middle of the distribution. They tended to have one to two fewer years of education than U.S.-born whites (1.4 fewer years for women and 2.2 fewer years for men). Finally, Southern and Central European immigrants had the lowest attainments, having about 3 to 4 fewer years of education than U.S.-born whites (Central Europeans: 2.9 fewer years for women, 2.7 fewer years for men; Southern Europeans: 3.6 fewer years for both women and men). These patterns were similar for both men and women.

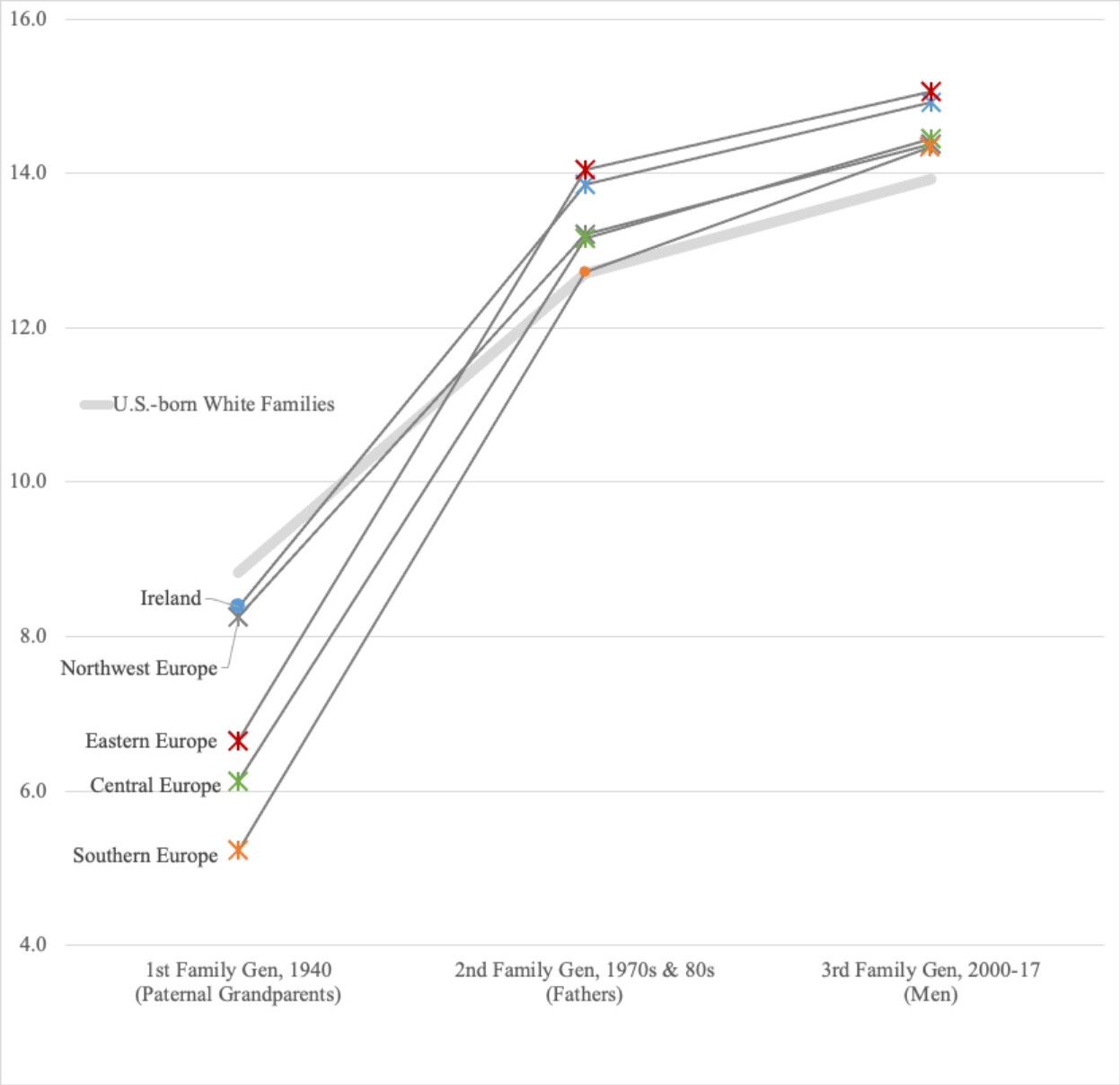
By the second family generation, the average educational attainment had increased substantially for both U.S.-born white and European immigrant families. This trend reflects the massive educational expansion that occurred during the post-war economic boom of the 1950s

Figure 2: Years of Education by Generational Status, Women (Matrilineal Side of Family)



Note: Significant differences between the European immigrant origin and U.S.-born white families are indicated by the "X"-shaped markers in the graphs (with small circles indicating nonsignificance).

Figure 3. Years of Education by Generational Status, Men (Patrilineal Side of Family)



*Significant differences between the European immigrant origin and U.S.-born white families are indicated by the "X"-shaped markers in the graphs (with small circles indicating nonsignificance).

and 1960s (Fischer and Hout 2006). More to the point, gaps in educational attainment between European immigrant and U.S.-born white families narrowed considerably. Among second generation women, only Southern and Central Europeans continued to have significantly lower levels of education than U.S.-born whites, but the gap had narrowed to less than a year of schooling. Additionally, the average educational attainment of Eastern European second generation women exceeded that of U.S.-born whites. Among second generation men, nearly all European origin groups had attainments that significantly exceeded those of U.S.-born whites; Southern Europeans were the exception, and they had nearly identical levels of education as U.S.-born whites.

By the third family generation, observed between 2000 and 2017, we see many reversals in group rankings. Whereas U.S.-born whites had the highest level of education in 1940, by the third generation, this group ranked lowest. Among men, *all* of the European origin groups had significantly higher levels of education than U.S.-born whites. Among women, the grandchildren of Southern, Central, and Eastern European immigrants had significantly higher levels of education than U.S.-born whites, while the grandchildren of Northwestern European and Irish women were not significantly different from them. Importantly, the groups at the bottom (Southern and Central Europeans) and middle (Eastern Europeans) of the distribution in 1940, had attained significantly more years of education than U.S.-born whites by the third generation among both men and women.

These patterns are robust to adjustments for demographic factors. Table 2 shows the coefficients for immigrant ethnic origin groups relative to U.S.-born whites in OLS models predicting years of education for the second and third family generations for women (Panel A) and men (Panel B). Model 1 excludes controls and Model 2 adjusts for year of birth, region of

Table 2: Educational Attainment by Ethnic Origin Among Second and Third Family Generation Women and Men (OLS Regression Models)

	Second Family Generation				Third Family Generation			
	Model 1		Model 2		Model 1		Model 2	
Panel A. Women								
Ethnic Origin (NH-White USB = reference)								
Northwest Europe	0.092		0.025		0.183		0.098	
Ireland	0.263		0.334	*	0.295		0.192	
Eastern Europe	0.458	**	0.480	**	1.065	***	0.959	***
Central Europe	-0.266	*	-0.054		0.398	***	0.319	**
Southern Europe	-0.526	***	-0.371	***	0.494	***	0.406	***
R-squared	0.004		0.083		0.007		0.029	
N	11,000				8,500			
Panel B. Men								
Ethnic Origin (NH-White USB = reference)								
Northwest Europe	0.491	***	0.432	***	0.450	***	0.327	**
Ireland	1.142	***	1.047	***	1.001	***	0.846	***
Eastern Europe	1.331	***	1.330	***	1.141	***	1.026	***
Central Europe	0.449	***	0.635	***	0.524	***	0.416	***
Southern Europe	0.015		0.131		0.420	***	0.306	*
R-squared	0.007		0.085		0.011		0.024	
N	12,500				9,000			

* p<0.05; ** p<0.01; ***p<0.001

Notes: IGENS-20 Sample, Vintage 1 (3-generation families; 1940 Census; 1973, 1979, 1981-1990 CPS; Census 2000; 2001-2017 ACS). Approved for disclosure on 11/2/20; review number: CBDRB-FY21-039. Model 1 is unadjusted for demographic controls. Model 2 controls for year of birth, region of residence in childhood (Northeast, Midwest, South, West), household size in childhood, and having two coresidential parents in childhood.

residence, household size and having two co-residential parents in childhood (coefficients for controls not shown; full models shown in Appendix Tables 1 and 2). In most cases, the level and significance of the coefficients is similar for Models 1 and 2. The only exceptions occured for second-generation women, among whom two European ethnic groups appeared more advantaged relative to U.S.-born whites after adjustments were made. The positive coefficient for Irish women increased and gained significance and the negative coefficient for Central European women attenuated toward zero and lost significance in Model 2.

Overall, the results show evidence of accelerated upward mobility among immigrant families. European-origin immigrant groups, including those with the lowest (e.g., Southern and Central Europeans) and mid-level (Eastern European) status in 1940, not only converged with U.S.-born whites but leapfrogged over them. This occurred as early as the second generation for Eastern European women and Eastern and Central European men, and by the third generation for nearly all groups.

Intergenerational Mobility

To what extent are group differences in educational attainment in the second and third generation attributable to differences in intergenerational mobility, as opposed to differences in parent and grandparent disadvantages? To answer this question, we added the educational attainment of prior generations to the models. Results are shown in Table 3 separately for women (Panel A) and men (Panel B). For reference, Model 2 is repeated from Table 2; it includes the ethnic origin dummy variables and demographic controls (coefficients for controls not shown; full models shown in Appendix Tables 1 and 2). For models predicting the educational attainment of second-generation parents, Model 3 adds the average educational attainment of the

Table 3: Educational Mobility Among Second and Third Family Generation Women and Men (OLS Regression Models)

	Second Family Generation				Third Family Generation					
	Model 2		Model 3		Model 2		Model 3		Model 4	
Panel A. Women										
Ethnic Origin (NH-White USB = reference)										
Northwest Europe	0.025		0.117		0.098		0.008		0.018	
Ireland	0.334	*	0.518	***	0.192		0.090		0.125	
Eastern Europe	0.480	**	1.025	***	0.959	***	0.602	***	0.693	***
Central Europe	-0.054		0.671	***	0.319	**	0.294	**	0.396	***
Southern Europe	-0.371	***	0.503	***	0.406	***	0.512	***	0.645	***
Parent and Grandparent Educational Attainment										
Parent Education			0.283	***			0.345	***	0.325	***
Grandparent Education									0.045	***
R-squared	0.083		0.197		0.029		0.177		0.179	
N	11,000		11,000		8,500		8,500		8,500	
Panel B. Men										
Ethnic Origin (NH-White USB = reference)										
Northwest Europe	0.432	***	0.652	***	0.327	**	0.256	**	0.279	**
Ireland	1.047	***	1.276	***	0.846	***	0.561	**	0.605	***
Eastern Europe	1.330	***	2.111	***	1.026	***	0.770	***	0.876	***
Central Europe	0.635	***	1.590	***	0.416	***	0.335	**	0.435	***
Southern Europe	0.131		1.421	***	0.306	*	0.394	***	0.517	***

Continued on next page.

Table 3 (Continued). Educational Mobility Among Second and Third Family Generation Women and Men (OLS Regression Models)

	Second Family Generation		Third Family Generation		
	Model 2	Model 3	Model 2	Model 3	Model 4
Parent and Grandparent Educational Attainment					
Parent Education		0.380 ***		0.394 ***	0.376 ***
Grandparent Education					0.043 ***
R-squared	0.085	0.202	0.024	0.195	0.197
N	12,500	12,500	9,000	9,000	9,000

* p<0.05; ** p<0.01; ***p<0.001

Notes: IGENS-20 Sample, Vintage 1 (3-generation families; 1940 Census; 1973, 1979, 1981-1990 CPS; Census 2000; 2001-2017 ACS). Approved for disclosure on 11/2/20; review number: CBDRB-FY21-039. All models control for year of birth, region of residence in childhood (Northeast, Midwest, South, West), household size in childhood, and having two coresidential parents in childhood (1=yes). First generation grandparent education is the average of first generation grandparents' education (maternal grandparents for second generation mothers, paternal grandparents for second generation fathers, and all grandparents for third generation women and men). Second generation parent education is the average of second generation mother's and father's education.

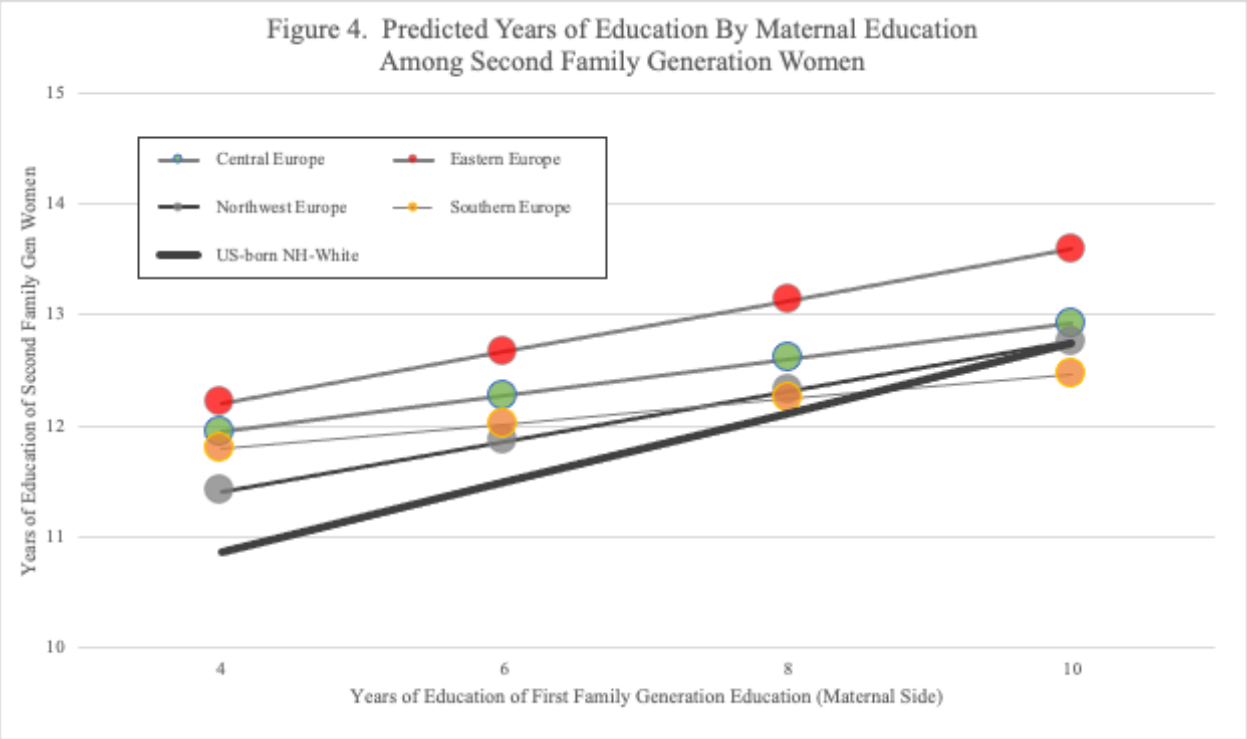
first family generation, or in other words, the parents of the second generation (maternal for women and paternal for men). For models predicting the educational attainment of third generation adults, Model 3 adds the average education of the second family generation, or the parents of the third generation, and Model 4 adds the average education of the first family generation, that is, the grandparents of the third generation (both maternal and paternal sides). .

We first focus on results for second-generation parents (first two columns of Table 3). Recall that Southern European women lagged behind their US.-born white counterparts, while other groups were either on par with or exceeded them (this is also reflected in Model 2). However, after we accounted for the significant effects of first family generation education (i.e., parents' education) in Model 3 (b = .283* for women and b = .380* for men), ethnic origin variations diminished and nearly all European origin groups attained significantly more years of schooling than their U.S.-born white counterparts. Groups with advantages in Model 2 showed even larger advantages in Model 3, and groups with disadvantages (Southern European women) or who were on par with U.S.-born whites (Central European women and Southern European men), joined the other groups by showing significant advantages relative to U.S.-born whites in Model 3. The only exception was that second-generation Northwestern European women did not differ significantly from U.S.-born whites regardless of whether first family generation education was accounted for. Overall, the results provide evidence that the children of immigrants experienced accelerated upward mobility relative to U.S.-born whites between the first and second generations. If given equal levels of parent education, all groups except northwest Europeans would have leapfrogged the attainments of U.S.-born whites by the second generation. Notably, this includes those who arrived with the lowest status: Southern and Central Europeans.

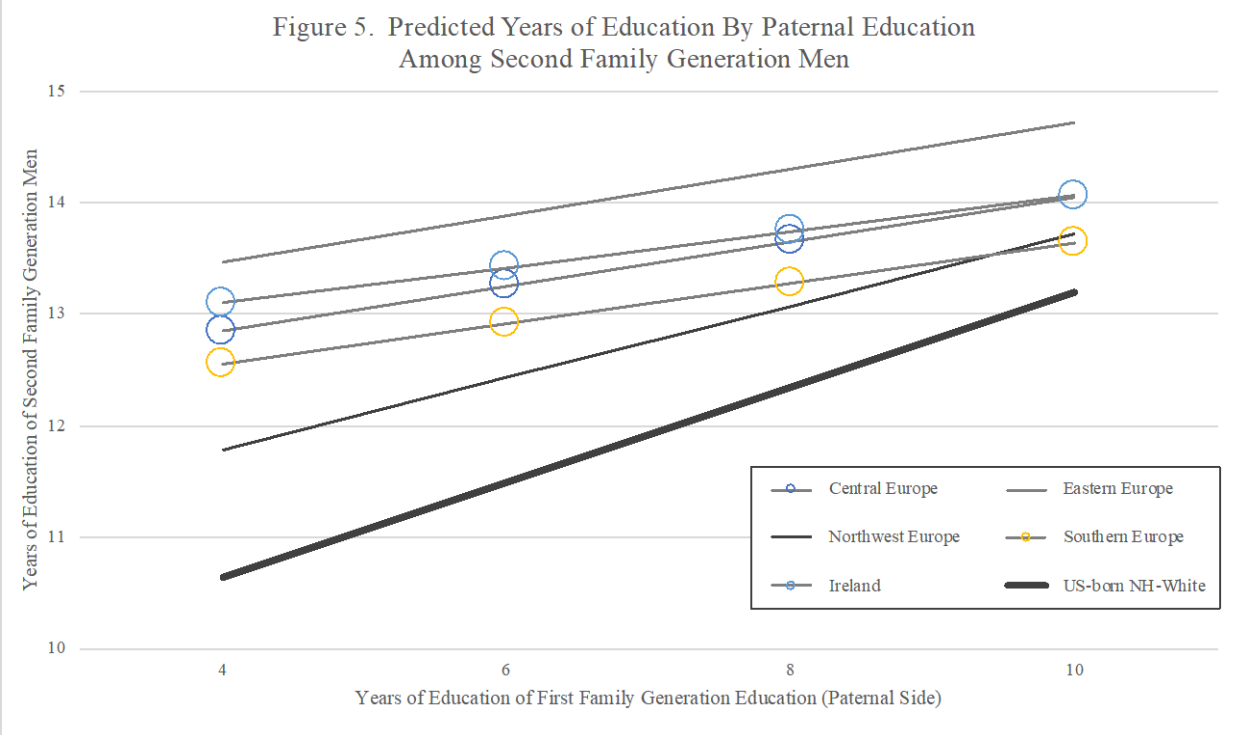
It is also noteworthy that the European immigrant-mobility advantage was larger for men than for women by half year of schooling or more (e.g., the advantage for Southern Europeans was .502 for women and 1.421 for men). Clogg tests suggest that these gender differences were significant for all groups, implying that, at least during the post-WWII era, the sons of immigrants attained higher levels of education relative to their U.S.-born white counterparts than the daughters of immigrants.

To explore the mobility of the second generation further, we tested interactions between parent education and ethnic origin, and they were highly statistically significant. Full model results are shown in Appendix Table 1. To facilitate interpretation, we graphed predicted values generated from these models in Figures 4 (women) and 5 (men). Figure 4 omits the Irish because their values were no different that the U.S.-born. Results indicate that the children of immigrants for most groups attained particularly high levels of education relative to U.S.-born whites when their parents had low levels of education. At higher levels of parental education, European-origin attainments were more similar to (yet still higher than) those of U.S.-born whites, among men; the same holds among women with the exception of the children of Southern European immigrant mothers. It is possible that for women in groups with already higher overall levels of attainment, the second generation had less room for improvement (as speculated by While and Mullen 2016), especially when their parents (the first generation) had high levels of education.

Overall, this pattern of accelerated upward mobility between the first and second generation is consistent with past research on contemporary children of immigrants. Labeled an "immigrant paradox" and attributed to unique family and community dyanmics among immigrants (i.e., due to cultural factors, selective migration, or ethnic social capital), this phenomenon is the primary reason European immigrant families were able to make up for their



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initial status disadvantages so quickly, often within a single generation.

More surprisingly, we found that the mobility advantage observed for the second generation extended into the third generation. As shown in the models estimated for third generation adults (last three columns in Table 3), parental education (Model 3) and grandparent education (Model 4) have significant and independent effects on third-generation educational attainment. Thus the expectations of the status attainment model for the importance of parental education and even grandparent education are borne out in these results. As shown in other work (Anderson, Sheppard, and Monden 2018), socioeconomic status (including educational attainment) of both parents and grandparents matter although the effects of grandparent attainment are only about one-eighth that of parental attainment. Yet we also see evidence of accelerated mobility between the second and third generation. The grandchildren of European-origin immigrants continued to display significant advantages of roughly half a year of schooling or more over their U.S.-born white counterparts, even after the educational attainment of the second and first generation (their parents and grandparents) were accounted for in Models 3 and 4. The only exceptions were women in the two highest-status immigrant groups from 1940: those whose grandparents immigrated from Northwest Europe or Ireland. The educational attainment of these two groups was not significantly different from U.S.-born whites after controlling for parent/grandparent attainment.

Also noteworthy is that the gender differences in the immigrant-mobility advantage in the third generation were not nearly as large as they were for the second generation. Compare, for example, the gender difference in the advantage for Southern Europeans. Among second-generation parents, men's advantage was nearly one year larger than women's (1.421 - .503 = .918). This difference reversed and was much smaller among the third generation adults (.517 -

.645 = -.128). Clogg tests confirm that gender differences in the ethnic origin coefficients in Model 4 were not significant for those of Eastern, Central, and Southern European origin. There continued to be significant male advantages among those of Northwestern Europe and Irish origin, however.

Finally, we tested interactions between parent education and origin group, but they were not significant (results not shown). This suggests that the advantages of being a grandchild of an European origin immigrant were similar regardless of class origins (i.e., educational attainment).

Robustness Tests

We conducted several anaylses in which we employed alternative coding schemes to assess the robustness of the findings. First, in our main analyses, we coded parent education and grandparent education as the average of up to two parents and four grandparents (taking information from all parents and grandparents that we could observe in our data). But in preliminary work, we also tested measures based on the highest level of education of the two parents or four grandparents, mother's/maternal grandmothers' education, and father's/paternal grandfathers' education, and results were similar to those presented here.

Additionally, in our main analyses, we defined ethnic origin for the third generation based on our combined measure of ethnic origin. The combined measure selects the ethnic origin of the person's grandparents (on both sides of the family) while giving priority to grandparents who had immigrant origins from certain European regions (i.e., in the order of Ireland, Northwest Europe, Southern Europe, Central Europe, and Eastern Europe). However, we recognize that this coding scheme could obscure alternative and equally valid ethnic origins. Therefore, we repeated the models for the third generation multiple times, using the matrilineal

and patrilineal versions of ethnic origin. Results for Model 4 are shown in Appendix Table 3. The coefficients for ethnic origin areconsistent across the different models. Additional models (e.g., Model 1, 2, and 3) are available upon request and they show the same level of consistency.

We also constrained our sample to assess whether period of arrival influenced our results. Due to the severe immigration restrictions brought by the Johnson Reed Act of 1924, it is possible that immigrants who arrived after this time period were more positively selected relative to industrial-era immigrants who arrived before this time period. Unfortunately, year of arrival is unavailable in the 1940 data. However, in supplemental analyses we restricted our sample to a group more likely to have arrived before 1924—those born before 1903. This yielded similar results to our main analyses. Even among the pre-1903 group, second-generation Southern, Central, and Eastern Europeans as well as the Irish experienced greater upward mobility than did their U.S.-born white contemporaries for both men and women. The only differences in the third generation were null findings for Eastern and Central Europeans for the women, and among Northwest Europeans for the men, denoting no difference in education (yet the coefficients are still relatively large and positive).

Another similar concern with sample selection is that the second generation in our sample is restricted to those who were living at home with one or both parents in 1940. Therefore, the sample excludes those who left the home early. This exclusion may bias the results if children who left home at younger ages are less upwardly mobile—and have lower educational attainments—than those who remained at home. To assess this possibility, we limited the 1940 sample to include children of immigrants who are aged 10 or younger. Very few children would have left the home by the age of 10. Again, we find significantly greater upward mobility between the first and second generation for most European origin groups except Irish men and

women as well as Eastern and Northwest European women. Both Central and Southern European men and women have higher educational attainments than U.S.-born whites in the second generation. These supplementary analyses are available upon request.

Discussion

Broad patterns gleaned from Census Bureau data over the course of the twentieth century overwhelmingly suggested rapid integration of second- and third-generation European ethnic groups that arrived in the United States during the early twentieth century. Yet limitations in these same data precluded addressing more fine-grained questions about rates of mobility from one generation to the next, ethnic variation in these rates, and the degree to which these groups were able to close initial gaps in attainments with U.S.-born whites. Different methods designed to work around these limitations in previous studies have yielded divergent empirical results–especially regarding the ethnic populations less proximate to the immigrant experience (i.e., the third (grandchild) generation and beyond)—leading to an empirical literature that is marked by an apparent lack of consensus. In the social sciences and the broader American discussion about immigration and diversity that is informed by social science research, this lack of empirical consensus appears to have provided an opening for arguments that immigrants, particularly those arriving with low skills, have not contributed to the vibrancy of the American economy and society (e.g., Brimelow 1997; Borjas 1994).

In this study, we utilize newly-available linkages of several censuses and surveys to bring additional clarity to questions about the integration of industrial-era European immigrants. We observe educational attainment across three family generations, something that has been historically difficult to capture due to limitations in data availability (Jimenez 2018). Our data

and methods make it possible to connect immigrants' socioeconomic status as measured by educational attainment with those of their children and grandchildren, without biases that owe to imprecise measurement of generational status or intergenerational changes in ethnic identity, thus providing a clearer assessment than has been available in prior research of the educational progress of both high- and low-skilled European immigrant groups who arrived around the turn of the twentieth century. We find that, not only did these immigrants catch up with their U.S.-born white contemporaries, most ethnic groups outpaced them. Noteably, this occurred quickly and even for those groups that were initially ranked at the bottom of the status hierarchy via their educational attainment and group origin status, such as Southern and Central Europeans. For many groups and especially among men, integration with and surpassing U.S.-born whites occurred as early as the second generation.

Consistent with prior research (Hirschman and Kraly 1990), we find that in 1940, most European immigrant groups had lower levels of education than U.S.-born whites. Central Europeans and Southern Europeans (made up primarily of people originating from the southern regions of Italy and Sicily) were particularly disadvantaged, having educational deficits of 2.5 to 3.5 years. Yet, by the second generation, most European immigrant groups had achieved educational attainments that matched or exceeded those of U.S.-born whites, and by the third generation, nearly all groups exceeded them. Although the results show some evidence of the ethnic "stickiness" that Borjas (1994) discussed, in that the lowest ranked groups among the first generation were also among the lowest ranked in the second generation *among the European immigrant groups*, the results also show considerable narrowing of differences among immigrant groups and between these groups and those of US.-born whites. Moreover, ethnic disadvantages did not persist beyond the second generation, and in many cases, group rankings in the first

generation were upended. For example, first generation Southern Europeans were the lowest performing immigrants in average years of education (and among the most highly discriminated against), yet by the third generation, Southern Europeans outperformed Northwest Europeans and U.S.-born whites, groups with the highest average levels of education, or "starting points", in 1940.

These changes in education came about because of the exceptional upward educational mobility among both the children and the grandchildren of immigrants. After adjusting for parents' and grandparents' education, nearly all (except Northwest European and Irish) industrial-era European second and third generation ethnics had attained higher levels of education than their U.S.-born white contemporaries. As shown in Figure 3, steep upward mobility was particularly evident for sons of immigrants whose parents had the lowest levels of education during the post-World War II era (i.e., those who attended school in the 1940s and 1950s). The preference of sons over daughters for investment in education may have been due to the greater opportunities for men compared with women in higher education and the professional workforce during the middle of the 20th century, but it may also be related to strategies enacted among immigrant families to invest scarce resources in their sons' education (Sassler 1995, 2006).

Our data do not permit us to analyze the precise reasons for the success of immigrants and their descendants, but we speculate that it may derive from any of a number of different factors, including the support and assistance of co-ethnic communities, pressure for children to succeed in immigrant families sometimes labeled the "immigrant bargain" in studies of contemporary immigrant families (Louie 2012), and European immigrants' opportunities for success given the educational expansions occurring in the Northeast during the post-WWII era.

The rapid upward trajectory from the bottom to the top of the status hierarchy of Southern European immigrants, a group composed primarily of impoverished tenant farmers in southern Italy and Sicily who occupied industrial and construction occupational niches in early twentieth century America, is particularly striking. As has been discussed in other work (Alba and Nee 2003), this group may have experienced a confluence of factors that led to their success, including strong family networks and a labor-migration orientation that reinforced obligations to support kin through hard work, a pattern that has perhaps not coincidentally also been observed for contemporary Mexican labor migrants (Van Hook and Bean 2009). Involvement in the Catholic school system, together with the expansion of educational opportunities for men following WWII, may have rerouted such work orientations from industrial and construction jobs into education, particularly among men.

The geographic concentration of industrial-era European immigrants in communities in the northeastern part of the United States may have also propelled their upward mobility. Although the results did not change when we added controls for geographic region, these indicators did not directly measure the degree of investment made in education at the local level. During the early twentieth century, the quality of schooling varied considerably within states, and European immigrants may have been concentrated in communities that made especially intensive investments in local schools. The children of industrial-era immigrants disproportionately came of age in Northeastern and Midwestern industrial centers. School systems in such jursidictions during the early decades of the twentieth century were at the forefront of the national effort to expand mass and post-secondary schooling, and boasted educational institutions robust enough to attenuate ethnoracial inequalities in accessing educational opportunities (Richardson, 1980, 1984). By contrast, U.S.-born-whites were more

heavily concentrated in the Northeast than industrial-era immigrants, where formal state education institutions were slow to develop. To the extent that such variation in schooling contexts is not accounted for by our geographic region controls, our coefficients representing ethnic difference may in part reflect unmeasured contextual variables, and therefore be biased, most likely in favor of children of industrial-era immigrants.

Another possibility is that the patterns we observe derive from selection into our sample. We already apply inverse probability treatment weights to adjust for biases related to the linkage of families across generations, and when we did this, the results did not change. Additionally, our supplementary analyses suggest that our results are robust to potential biases due to the inclusion of immigrant families who arrived following the passage of the Johnson-Reed Act in 1924 and to the absence of second-generation children who had already left their parental home by 1940. But other biases may exist. For example, we are only able to observe immigrants who moved to the United States and remained there. Between one-third and one-half of European immigrants returned to their countries of origin, and it is possible that only those who were successful in the United States remained there (Abramitzky et al. 2017; Gabaccia 2013). Also, Feliciano and Lanuza (2017) point to selection into immigration as key to understanding children of immigrants' exceptional upward mobility. They argue that even if immigrants' educational attainments are low relative to the U.S.-born, immigrants tend to have relatively high levels of education compared with nonimmigrants in their home country. Thus, although their children experience upward mobility in absolute terms, the second generation does not experience upward mobility in relative terms; they merely reproduce their parents' relative class position. We suspect that this explanation may at least partially underlie the rapid upward mobility of the children and grandchildren of Eastern European immigrants, a group that was composed mostly

of Jewish immigrants and who had unusually high literacy rates compared with other Europeans. It is unclear, however, that Feliciano and Lanuza's explanation applies to Southern European immigrants. As noted above, they originated from the lower rungs of society in southern Italy and Sicily in the late 1800s and early 1900s, which was itself among the poorest regions of Europe at the time. Poverty did not prevent working-age men from emigrating but most likely propelled them to do so, and many, at least initially, financed their trips through a contract labor system whereby they would be obligated to repay their *padrone* for steamship tickets and connections to jobs after they arrived in their destination (Gabaccia 2013).

We caution readers that the results presented here about European immigrants may not apply to contemporary immigrants. For example, while there are some important similarities between Mexican and Southern European immigrants (both exhibit high levels of familism, engage in circular labor migration, have comparatively low levels of education, and experienced discrimination), there are also some important differences, such as the burdens placed on Mexicans related to the large share with undocumented legal status, and the limited educational opportunities for Mexican Americans in the Southwest before the civil rights movements of the 1960s (Perlmann 2005). It thus remains unknown whether immigrants who may experience even greater structural and interpersonal discrimination than did even the least advantaged European origin groups, such as Mexicans, would fare worse.

Exploring all of these issues extends beyond the scope of this paper, but as of now, the results provide evidence of "leapfrogging" among European immigrant groups over U.S.-born whites. This pattern, in which the lowest-status groups rose to become the highest-status groups within three generations, is clearly inconsistent with the idea that immigrant groups with low skills have more difficulty integrating into the American mainstream. It is even somewhat

inconsistent with the “melting pot” model of integration. Instead, it supports an “immigrant exceptionalism” argument, whereby immigrants tended to quickly erase their initial deficits in skills and education by the second generation and even exceed those of the U.S.-born “mainstream” by the third. For many decades, immigration restrictionists have worried about the American practice of admitting low-skilled immigrants on the basis of their family connections to the United States, and some of them pointed to industrial-era immigrants as evidence for their arguments. Our research helps put to rest some of these concerns about early-arriving European immigrants. Future research should evaluate the extent to which our findings hold across generations for contemporary immigrant groups.

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Appendix Table 1: OLS Models Predicting Years of Education Among Mothers and Fathers in the Second Family Generation

	Second Family Generation Women				Second Family Generation Men			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
Ethnic Origin (NH-White USB = reference)								
Northwest Europe	0.092	0.025	0.117	0.897**	0.491***	0.432***	0.652***	1.583***
Ireland	0.263	0.334*	0.518***	1.437**	1.142***	1.047***	1.276***	3.522***
Eastern Europe	0.458**	0.480**	1.025***	1.659***	1.331***	1.330***	2.111***	3.681***
Central Europe	-0.266*	-0.054	0.671***	1.670***	0.449***	0.635***	1.590***	3.124***
Southern Europe	-0.526***	-0.371***	0.503***	1.726***	0.015	0.131	1.421***	2.898***
Class Origins (Educational Attainment)								
First Family Generation			0.283***	0.311***			0.380***	0.425***
Origin * First Family Generation Education								
Northwest Europe				-0.087**				-0.107**
Ireland				-0.108				-0.266***
Eastern Europe				-0.081*				-0.215***
Central Europe				-0.149***				-0.227***
Southern Europe				-0.199***				-0.245***
Year of birth		0.027***	0.009*	0.011**		0.061***	0.035***	0.038***
Region in Childhood (Northeast = reference)								
Midwest		-0.07	-0.087	-0.073		-0.230***	-0.242***	-0.222***
South		-0.510***	-0.226***	-0.182**		-0.727***	-0.351***	-0.284***
West		0.462***	0.225**	0.230**		0.521***	0.224*	0.217*
Household Size in Childhood		-0.245***	-0.153***	-0.152***		-0.271***	-0.166***	-0.162***
Two parents		0.298***	0.242**	0.253**		0.704***	0.495***	0.481***
CPS year		0.024***	0.024***	0.024***		0.015**	0.013**	0.011*

Continued on next page.

Appendix Table 1 (Continued): OLS Models Predicting Years of Education Among Mothers and Fathers in the Second Family Generation

	Second Family Generation Women				Second Family Generation Men			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
Constant	12.41***	-85.53***	-54.98***	-57.40***	12.70***	-134.6***	-82.25***	-86.28***
R-squared	0.004	0.083	0.197	0.202	0.007	0.085	0.202	0.209
N	11,000				12,500			

* p<0.05; ** p<0.01; ***p<0.001

Notes: IGENS-20 Sample, Vintage 1 (3-generation families; 1940 Census; 1973, 1979, 1981-1990 CPS; Census 2000; 2001-2017 ACS). Approved for disclosure on 11/2/20; review number: CBDRB-FY21-039. First family generation education is averaged across both maternal grandparents for mothers and paternal grandparents for fathers.

Appendix Table 2: OLS Models Predicting Years of Education Among Women and Men in the Third Family Generation

	Third Family Generation Women				Third Family Generation Men			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
Origin (NH-White USB = reference)								
Northwest Europe	0.183	0.098	0.008	0.018	0.450***	0.327**	0.256**	0.279**
Ireland	0.295	0.192	0.09	0.125	1.001***	0.846***	0.561**	0.605***
Eastern Europe	1.065***	0.959***	0.602***	0.693***	1.141***	1.026***	0.770***	0.876***
Central Europe	0.398***	0.319**	0.294**	0.396***	0.524***	0.416***	0.335**	0.435***
Southern Europe	0.494***	0.406***	0.512***	0.645***	0.420***	0.306*	0.394***	0.517***
Class Origins								
Second Family Generation			0.345***	0.325***			0.394***	0.376***
First Family Generation				0.045***				0.043***
Year of birth		0.043***	0.023***	0.022***		0.029***	0.006	0.005
Region (Northwest = reference)								
Midwest		-0.111	-0.027	-0.03		-0.086	-0.009	-0.011
South		0.406***	-0.159*	-0.133		0.476***	0.239***	-0.207**
West		-0.218*	0.392***	-0.415***		-0.042	-0.219*	-0.242**
Household Size in Childhood		0.109***	0.078***	-0.078***		-0.051**	-0.015	-0.015
Two parents		0.282***	0.196***	0.188***		0.301***	0.213***	0.200***
Constant	14.15***	69.13***	34.27***	-32.17***	13.83***	42.25***	-2.787	-0.366
R-squared	0.007	0.029	0.177	0.179	0.011	0.024	0.195	0.197
N	8,500				9,000			

* p<0.05; ** p<0.01; ***p<0.001

Notes: IGENS-20 Sample, Vintage 1 (3-generation families; 1940 Census; 1973, 1979, 1981-1990 CPS; Census 2000; 2001-2017 ACS). Approved for disclosure on 11/2/20; review number: CBDRB-FY21-039. First family generation education is averaged across all grandparents; second family generation education is averaged across both parents.

Appendix Table 3: Sensitivity of OLS Models of Third Family Generation's Educational Attainment to Coding of Origin (Combined Matrilineal & Patrilineal, Matrilineal, or Partrilineal Side of the Family)

	Third Family Generation Women			Third Family Generation Men		
	Combined	Matrilineal	Patrilineal	Combined	Matrilineal	Patrilineal
Northwest Europe	0.018	0.060	-0.060	0.279**	0.249*	0.302**
Ireland	0.125	-0.143	0.379	0.605***	0.468*	0.797***
Eastern Europe	0.693***	0.499*	0.685**	0.876***	0.950***	0.687***
Central Europe	0.396***	0.355*	0.414**	0.435***	0.345*	0.563***
Southern Europe	0.645***	0.703***	0.484***	0.517***	0.503**	0.571***
N		8500			9000	

* p<0.05; ** p<0.01; ***p<0.001

Notes: IGENS-20 Sample, Vintage 1 (3-generation families; 1940 Census; 1973, 1979, 1981-1990 CPS; Census 2000; 2001-2017 ACS). Approved for disclosure on 11/2/20; review number: CBDRB-FY21-039. Models control for all factors in Model 3 of Table 3.